

Ex.No.8 Use Steg-Expose to detect hidden data in images

Description

StegExpose is a tool used to detect hidden data in images using steganography analysis. It works by evaluating the statistical properties of an image to estimate whether any hidden data is embedded. Below is a step-by-step guide on how to use StegExpose to detect hidden data in images:

Prerequisites:

1. **Java Runtime Environment (JRE):** StegExpose is a Java-based tool, so ensure you have JRE installed.
2. **StegExpose Tool:** Download StegExpose from the [official GitHub repository](#).

Step-by-Step Process:

1. Download and Set Up StegExpose:

- **Download the tool:** Go to [StegExpose GitHub](#) and download the .jar file.
- **Install Java:** Ensure that Java is installed on your machine. You can download it from the official site.
- **Prepare environment:** Place the .jar file in a folder where you plan to work on the image files.

2. Select Images for Analysis:

- Collect the images you suspect might have hidden data. StegExpose supports formats such as .png, .jpg, .bmp, etc.

3. Open Command Line or Terminal:

- Navigate to the folder where you have saved the StegExpose.jar file.
- Open your **Command Prompt** (Windows) or **Terminal** (Linux/macOS).

4. Run StegExpose on an Image:

- Use the following command to check a single image for hidden data:

```
bash
```

```
java -jar StegExpose.jar <image_file_path>
```

Example:

```
bash
```

```
java -jar StegExpose.jar test_image.png
```

- Replace <image_file_path> with the actual path of the image you want to check.

5. Analyze the Output:

- After running the tool, it will analyze the image and provide a score.
- **Output Explanation:**
 - StegExpose calculates a "suspect" score (between 0 and 1). The higher the score, the more likely that hidden data is present.
 - **Threshold values** (suggested by the tool):
 - **Less than 0.2:** Image is clean (no hidden data).

- **0.2 - 0.3:** Possibly some hidden data.
- **Above 0.3:** Likely that steganography is present.

6. Batch Analysis (Multiple Images):

- If you have multiple images to check, you can analyze them in bulk by specifying a folder. For example:

```
bash
```

```
java -jar StegExpose.jar <folder_path>
```

- Replace <folder_path> with the path to the folder containing the images.

7. Advanced Options (Optional):

- StegExpose provides additional parameters, such as adjusting the sensitivity of detection and output verbosity. Use --help to see the list of options:

```
bash
```

```
java -jar StegExpose.jar --help
```

8. Review the Results:

- For each image analyzed, review the scores and determine whether you need to investigate further.

Example:

```
bash
```

```
java -jar StegExpose.jar suspect_image.png
```

Output:

```
makefile
```

```
Analyzing suspect_image.png...
```

```
Result: 0.4
```

```
Steganography likely present
```

This output indicates that hidden data is likely present in the image suspect_image.png.

OUTPUTS:

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\anive>C:\platform-tools
'C:\platform-tools' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\anive>dir C:\platform-tools
Volume in drive C is Windows
Volume Serial Number is 1060-CD90

Directory of C:\platform-tools

20-09-2026 16:58 <DIR>
20-09-2026 16:45      8,273,560 adb.exe
20-09-2026 16:45      198,184 AdbWinApi.dll
20-09-2026 16:45      73,368 AdbWinUsbApi.dll
20-09-2026 16:45      451,224 etcdtool.exe
20-09-2026 16:45      2,420,080 fastboot.exe
20-09-2026 16:45      52,888 hprof-conv.exe
20-09-2026 16:45      243,016 libwinpthread-1.dll
20-09-2026 16:45      473,752 make_f2fs.exe
20-09-2026 16:45      473,752 make_f2fs_casefold.exe
20-09-2026 16:45      1,157 mke2fs.conf
20-09-2026 16:45      763,544 mke2fs.exe
20-09-2026 16:45      1,154,131 NOTICE.txt
20-09-2026 16:45      38 source.properties
20-09-2026 16:45      3,031,704 sqlite3.exe
                14 File(s)      17,529,398 bytes
                1 Dir(s)      239,791,525,888 bytes free

C:\Users\anive>|
```

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\anive> adb pair 10.1.4.145:37711
Enter pairing code: 856522
Successfully paired to 10.1.4.145:37711 [guid=adb-RZCX50T016B-6uhg61]

C:\Users\anive> adb devices
List of devices attached
adb-RZCX50T016B-6uhg61_adb-tls-connect.tcp device

C:\Users\anive>mkdir C:\forensics-output
A subdirectory or file C:\forensics-output already exists.

C:\Users\anive>cd C:\forensics-output

C:\forensics-output>adb shell content query --uri content://call_log/calls > callog.txt

C:\forensics-output> callog.txt

C:\forensics-output>adb shell content query --uri content://contacts/phones > contacts.txt
adb shell content query --uri content://sms > sms.txt
C:\forensics-output>
C:\forensics-output>contacts.txt

C:\forensics-output>sms.txt
'sms.txt' is not recognized as an internal or external command,
operable program or batch file.

C:\forensics-output>adb shell content query --uri content://sms > sms.txt

C:\forensics-output>adb shell content query --uri content://contacts/phones > contacts.txt

C:\forensics-output>contacts.txt

C:\forensics-output>sms.txt

C:\forensics-output>'adb disconnect'
'adb' is not recognized as an internal or external command,
operable program or batch file.
```

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\anive>java -version
openjdk version "25.0.3" 2026-04-21 LTS
OpenJDK Runtime Environment Temurin-25.0.3+9 (build 25.0.3+9-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.3+9 (build 25.0.3+9-LTS, mixed mode, sharing)

C:\Users\anive> cd Downloads

C:\Users\anive\Downloads>dir
Volume in drive C is Windows
Volume Serial Number is 1060-CD90

Directory of C:\Users\anive\Downloads

20-09-2026 21:43 <DIR> .
20-09-2026 16:59 <DIR> ..
06-09-2026 12:54      9,956 # IEEE Conference Paper Blueprint T.txt
23-06-2025 10:13      13,322 #Stich #Aesthetic.jpg
06-08-2025 20:31      704 -lock.matlab_R2025a_Windows (1).exe#
12-09-2025 15:49      22,017 1.html
19-09-2025 13:23      66,119 1.jpg
08-08-2025 16:25      104,169 11e4d8ec-69b1-42fd-b068-87b1f7a86bb3.jpeg
06-09-2026 11:10      859,934 132219.pdf
06-09-2026 10:59      286,413 14942-ArticleText-29134-1-10-20241209 (1).pdf
17-09-2026 07:31      25,923 1543562709_courseendproject_healthcareinsurancesolution.docx
17-09-2026 07:30      23,256 1543562880_courseendproject_identityverificationsolution.docx
16-09-2026 22:21      22,369 1556184157_lessonendproject_lesson2solution.docx
16-09-2026 23:08      14,273 1556184386_lessonendprojectlesson3solution.docx
17-09-2026 00:19      14,968 1556184878_lessonendprojectlesson4solution.docx
17-09-2026 06:30      14,972 1556185170_lessonendprojectlesson6solution.docx
17-09-2026 07:08      15,624 1556185393_lessonendprojectlesson7solution.docx
16-09-2026 12:14      14,115 1556185495_lessonendprojectlesson1solution (1).docx
16-09-2026 12:13      14,115 1556185495_lessonendprojectlesson1solution.docx
17-09-2026 01:01      13,727 1556185604_lessonendprojectlesson5solution.docx
17-09-2026 07:30      11,362 1556281664_practice1 (1).zip
17-09-2026 07:29      11,892 1556281664_practice1.zip
17-09-2026 07:30      60,430 1556281766_practice2.zip
17-09-2026 07:30      13,014 1556281828_practice3.zip
17-09-2026 07:30      169,621 1556281940_practice4.zip
```

```

C:\Users\anive\Downloads>cd C:\Users\anive\Downloads\StegExpose-master
C:\Users\anive\Downloads\StegExpose-master>dir
Volume in drive C is Windows
Volume Serial Number is 1060-CD90

Directory of C:\Users\anive\Downloads\StegExpose-master

20-09-2026  21:42    <DIR>          .
20-09-2026  21:43    <DIR>          ..
20-09-2026  21:43    <DIR>          StegExpose-master
                0 File(s)                0 bytes
                3 Dir(s)  243,009,679,360 bytes free

C:\Users\anive\Downloads\StegExpose-master>java -jar StegExpose.jar testFolder
Error: Unable to access jarfile StegExpose.jar

C:\Users\anive\Downloads\StegExpose-master>cd StegExpose-master
C:\Users\anive\Downloads\StegExpose-master\StegExpose-master>dir
Volume in drive C is Windows
Volume Serial Number is 1060-CD90

```

```

Directory of C:\Users\anive\Downloads\StegExpose-master\StegExpose-master

20-09-2026  21:43    <DIR>          .
20-09-2026  21:42    <DIR>          ..
20-09-2026  21:43             4,690 ChiSquare.class
20-09-2026  21:43             9,188 ChiSquare.java
20-09-2026  21:43      1,599,627 commons-math3-3.1.1.jar
20-09-2026  21:43      682,680 dissertation.pdf
20-09-2026  21:43             744 Fuse.class
20-09-2026  21:43             843 Fuse.java
20-09-2026  21:43      907 ImageFileManager.class
20-09-2026  21:43     1,902 ImageFileManager.java
20-09-2026  21:43             85 manifest.mf
20-09-2026  21:43      377 PixelBenchmark.class
20-09-2026  21:43     1,818 PixelBenchmark.java
20-09-2026  21:43     1,371 PrimarySets.class
20-09-2026  21:43     2,403 PrimarySets.java
20-09-2026  21:43      4,657 README.md
20-09-2026  21:43     4,730 README.md~
20-09-2026  21:43      83,762 roc.png
20-09-2026  21:43      8,959 RSAnalysis.class
20-09-2026  21:43     20,745 RSAnalysis.java
20-09-2026  21:43      5,202 RunStegExpose.class
20-09-2026  21:43     6,822 RunStegExpose.java
20-09-2026  21:43      3,322 SamplePairs.class
20-09-2026  21:43     6,348 SamplePairs.java
20-09-2026  21:43    10,621,897 StegExpose.analysis.ods
20-09-2026  21:43     3,060,550 StegExpose.jar
20-09-2026  21:43    <DIR>          testFolder
                24 File(s)        16,133,629 bytes
                3 Dir(s)  243,008,266,240 bytes free

C:\Users\anive\Downloads\StegExpose-master\StegExpose-master>java -jar StegExpose.jar testFolder
stego.6666458261_e455d262b5.z.png is suspicious. Approximate amount of hidden data is 114785 bytes.
stego.6672108499_85c582a7f9.png is suspicious. Approximate amount of hidden data is 137047 bytes.
stego.6672542201_532f70bffe.png is suspicious. Approximate amount of hidden data is 67141 bytes.

```

```

Directory of C:\Users\anive\Downloads\StegExpose-master\StegExpose-master\testFolder

20-09-2026  21:43    <DIR>          .
20-09-2026  21:43    <DIR>          ..
20-09-2026  21:43      396,683 clean_7221186570_82a10bc040.z.png
20-09-2026  21:43      133,396 clean_7222762386_2688184c8e.png
20-09-2026  21:43      403,907 clean_7228374452_8f743dc3d4.z.png
20-09-2026  21:43      585,648 clean_7228718722_1cf25dff3e.z.png
20-09-2026  21:43      561,078 clean_7231623538_f51db2a35a.z.png
20-09-2026  21:43      584,044 clean_7232193260_c2fd8c0f25.z.png
20-09-2026  21:43      565,315 clean_7232202430_d38b6d6986.z.png
20-09-2026  21:43      532,090 clean_7232206610_b8cfed120.z.png
20-09-2026  21:43      552,283 clean_7232206662_3d42c69109.z.png
20-09-2026  21:43      228,747 clean_7234473324_bb8a82b5bd.png
20-09-2026  21:43      348,412 clean_7235558256_3099066753.png
20-09-2026  21:43      340,760 clean_7235972310_8c25258da5.png
20-09-2026  21:43      228,650 clean_7237798856_86870c8823.z.png
20-09-2026  21:43      673,018 stego.6666458261_e455d262b5.z.png
20-09-2026  21:43      439,754 stego.6672108499_85c582a7f9.png
20-09-2026  21:43      367,778 stego.6672542201_532f70bffe.png
                16 File(s)         6,943,563 bytes
                2 Dir(s)  243,006,373,888 bytes free

C:\Users\anive\Downloads\StegExpose-master\StegExpose-master> mkdir myImages

C:\Users\anive\Downloads\StegExpose-master\StegExpose-master>java -jar StegExpose.jar myImages
stego.6666458261_e455d262b5.z.png is suspicious. Approximate amount of hidden data is 114785 bytes.
stego.6672108499_85c582a7f9.png is suspicious. Approximate amount of hidden data is 137047 bytes.
stego.6672542201_532f70bffe.png is suspicious. Approximate amount of hidden data is 67141 bytes.

```

```

C:\Users\anive\Downloads\StegExpose-master\StegExpose-master>notepad myResults_final.csv

```

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
	File Name	Suspicious?	Est. Hidden Data (bytes)	Primary Sets	Chi Square	Sample Pairs	RS Analysis	Fusion (mean)							
1															
2	clean_7221196570_82a10bc040_z.png	No	-	0.1133	0.1424	N/A	0.0487	0.1015							
3	clean_7222762386_2688184c9e.png	No	-	N/A	0	0.2401	0.2586	0.1662							
4	clean_7228374452_8f743dc344_z.png	No	-	0.0311	0.0074	N/A	0.0161	0.0182							
5	clean_7228718722_1cf25dff3e_z.png	No	-	0.1161	0.0021	N/A	0.0999	0.0727							
6	clean_7231623538_151db2a35a_z.png	No	-	0.0509	0.0189	N/A	0.0617	0.0438							
7	clean_7232193260_c28d8c0f25_z.png	No	-	0.2574	0.0027	0.0969	0.1068	0.1159							
8	clean_7232202430_d38b6d6986_z.png	No	-	N/A	0.0097	0.1802	0.2101	0.1333							
9	clean_7232206610_b8cfcd120_z.png	No	-	0.0285	0.036	0.0805	0.0844	0.0573							
10	clean_723220662_3d42cd9109_z.png	No	-	N/A	0.0125	0.1329	0.1474	0.0976							
11	clean_7234473324_bb8a82b5bd.png	No	-	0.0165	0.0021	N/A	0.0366	0.0184							
12	clean_7235558256_3099066753.png	No	-	0.1755	0.1769	N/A	0.2017	0.1847							
13	clean_7235972310_8c25258da5.png	No	-	0.185	0.0052	N/A	0.0973	0.0958							
14	clean_7237798856_86870c8823_z.png	No	-	0.0733	0.0009	N/A	0.0697	0.0479							
15	stego_6666458261_e455d262b5_z.png	Yes	114785	N/A	0.0677	0.7293	0.738	0.5117							
16	stego_6672108499_85c582a7f0.png	Yes	137047	N/A	1	N/A	0.8699	0.9349							
17	stego_6672542201_532f70bffe.png	Yes	67141	N/A	0.0937	0.7812	0.7681	0.5477							
18															